

HONG KONG EXAMINATIONS AND ASSESSMENT AUTHORITY
HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION 2016

BIOLOGY PAPER 1

8.30 am – 11.00 am (2 hours 30 minutes)

This paper must be answered in English

GENERAL INSTRUCTIONS

- (1) There are **TWO** sections, A and B, in this Paper. You are advised to finish Section A in about 35 minutes.
- (2) Section A consists of multiple-choice questions in this question paper. Section B contains conventional questions printed separately in Question-Answer Book B.
- (3) Answers to Section A should be marked on the Multiple-choice Answer Sheet while answers to Section B should be written in the spaces provided in Question-Answer Book B. **The Answer Sheet for Section A and the Question-Answer Book B for Section B will be collected separately at the end of the examination.**

INSTRUCTIONS FOR SECTION A (MULTIPLE-CHOICE QUESTIONS)

- (1) Read carefully the instructions on the Answer Sheet. After the announcement of the start of the examination, you should first stick a barcode label and insert the information required in the spaces provided. No extra time will be given for sticking on the barcode label after the 'Time is up' announcement.
- (2) When told to open this book, you should check that all the questions are there. Look for the words '**END OF SECTION A**' after the last question.
- (3) All questions carry equal marks.
- (4) **ANSWER ALL QUESTIONS.** You are advised to use an HB pencil to mark all the answers on the Answer Sheet, so that wrong marks can be completely erased with a clean rubber. You must mark the answers clearly; otherwise you will lose marks if the answers cannot be captured.
- (5) You should mark only **ONE** answer for each question. If you mark more than one answer, you will receive **NO MARKS** for that question.
- (6) No marks will be deducted for wrong answers.

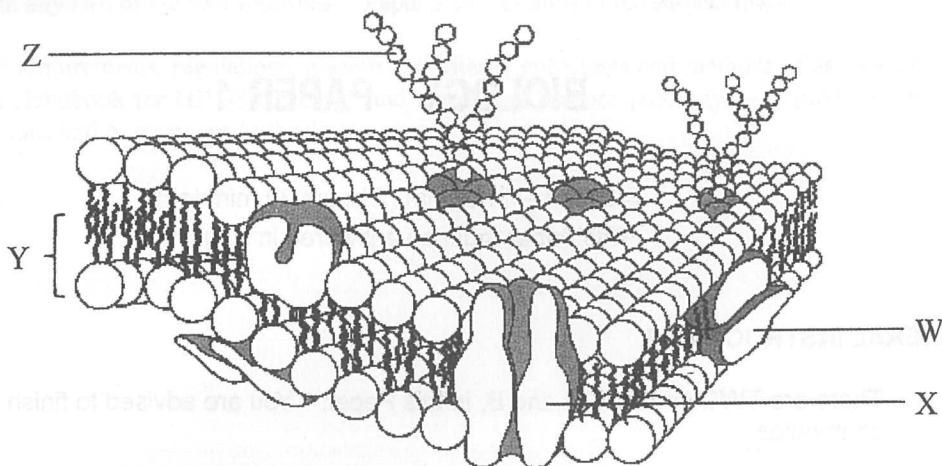
Not to be taken away before the end of the examination session

There are 36 questions in this section.

The diagrams in this section are NOT necessarily drawn to scale.

Directions:

Questions 1 and 2 refer to the schematic diagram below, which shows the structure of a cell membrane:



1. Which of the following molecules contributes to the fluidity of the cell membrane?

- A. W
- B. X
- C. Y
- D. Z

2. If the membrane is located at the inner surface of the small intestine, the absorption of amino acids is likely to be achieved via

- A. W.
- B. X.
- C. Y.
- D. Z.

3. In which of the following pairs of carbohydrates can Benedict's test be used to distinguish the two carbohydrates from one another?

- (1) sucrose and starch
- (2) sucrose and maltose
- (3) glucose and maltose
- (4) glucose and starch

- A. (1) and (3) only
- B. (1) and (4) only
- C. (2) and (3) only
- D. (2) and (4) only

4. Which of the following parts of the nucleotide make up the backbone of a polynucleotide strand?

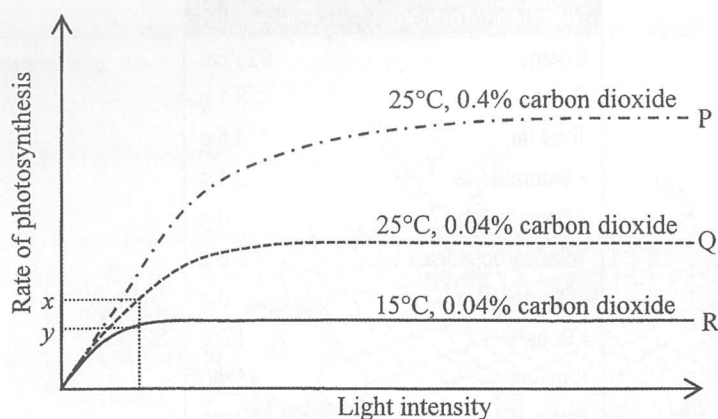
- A. sugar and base
- B. sugar and phosphate
- C. base and phosphate
- D. base, sugar and phosphate

Directions: Questions 5 and 6 refer to the diagram below, which shows the nutrition label of a food product:

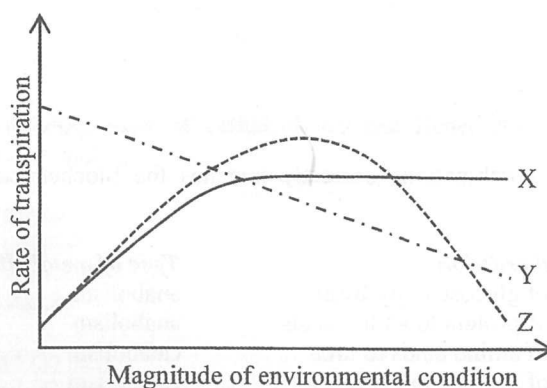
Nutrition Information	Per serving
Energy	62 Kcal
Protein	3.1 g
Total fat	3.5 g
- Saturated fat	2.4 g
- Trans fat	0 g
Total carbohydrate	4.6 g
- Dietary fibre	0 g
- Sugars	4.6 g
Sodium	44 mg
Calcium	110 mg

5. Which of the following food substances provides most of the energy content in this food product?
- fat
 - protein
 - sodium
 - carbohydrate
6. The food product bearing this nutrition label is most likely to be
- milk.
 - bread.
 - sausages.
 - potato chips.
7. Which of the following organs produces digestive juice that **does not** contain enzymes?
- liver
 - mouth
 - stomach
 - pancreas
8. Which of the following combinations correctly matches the biochemical reaction with the type of metabolism involved?
- | | Biochemical reaction | Type of metabolism |
|----|--------------------------------------|---------------------------|
| A. | conversion of glucose to pyruvate | anabolism |
| B. | conversion of protein to amino acids | anabolism |
| C. | conversion of amino acids to urea | catabolism |
| D. | conversion of glucose to glycogen | catabolism |

Directions: Questions 9 and 10 refer to the diagram below, which shows the rate of photosynthesis of a plant under different conditions:



9. The factor(s) that determine(s) the difference in the values of *x* and *y* in the diagram is / are
- temperature.
 - light intensity.
 - carbon dioxide concentration.
 - light intensity and temperature.
10. Which of the following descriptions best accounts for the higher rate of photosynthesis of P as compared to Q?
- All conditions are optimum in P.
 - There is a faster diffusion of carbon dioxide into the leaves.
 - More carbon dioxide can be used as raw material for photosynthesis.
 - Carbon dioxide concentration in P is much higher than that in the atmosphere.
11. The graph below shows the changes of the transpiration rate of a plant under different environmental conditions:



Which of the following correctly matches the environmental conditions represented by X, Y and Z?

- | | X | Y | Z |
|----|-------------------|-------------------|-------------------|
| A. | light intensity | wind speed | relative humidity |
| B. | wind speed | light intensity | relative humidity |
| C. | relative humidity | light intensity | wind speed |
| D. | light intensity | relative humidity | wind speed |

12. A student wants to estimate the stomatal density of the upper and lower epidermis of a leaf using a microscope. Which of the following combinations correctly shows the magnification that should be used and the reason?

	Magnification	Reason
A.	low magnification	show a larger field of view
B.	low magnification	give a brighter image
C.	high magnification	show more cellular details
D.	high magnification	give a clearer image

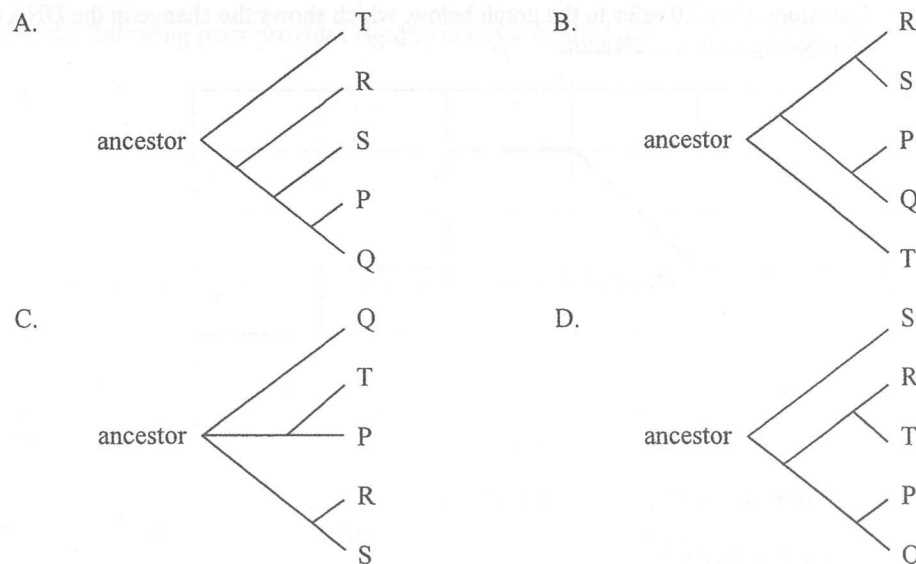
13. Of the following blood vessels, which one contains blood with the highest urea content?

- A. renal vein
- B. hepatic vein
- C. pulmonary vein
- D. hepatic portal vein

Directions: Questions 14 and 15 refer to the information below. Five new species of eubacteria were discovered in Antarctic ice core samples. The nucleotide sequences of the gene that codes for an essential protein of these new species were compared. The table below shows the number of nucleotide differences between the species:

Species	Number of nucleotide differences				
	P	Q	R	S	T
P	-	4	12	11	22
Q		-	12	11	19
R			-	4	22
S				-	22
T					-

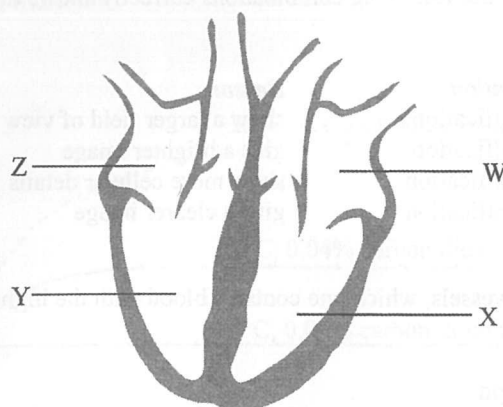
14. Which of the following evolutionary trees best illustrates the phylogenetic relationship of the five species?



15. Which of the following cell components can be found in these species?

- A. nucleus
- B. cell wall
- C. chloroplast
- D. mitochondrion

Directions: Questions 16 and 17 refer to the diagram below, which shows a section of the heart:



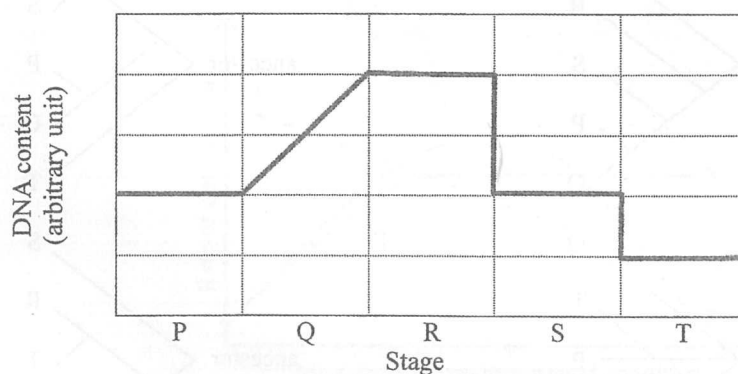
16. When W contracts, which of the following combinations about the conditions of the valves is correct?

	<i>Semi-lunar valve</i>	<i>Bicuspid valve</i>
A.	open	open
B.	open	closed
C.	closed	open
D.	closed	closed

17. Which of the following statements about the heart is correct?

- Z receives oxygenated blood from the lungs.
- Y and Z pump out same volume of blood.
- Blood in Z has higher glucose content than that of W.
- Blood pumped out from Y travels a longer distance than that of X before returning to the heart.

Directions: Questions 18 to 20 refer to the graph below, which shows the change in the DNA content of a cell undergoing a certain division:



18. Which of the following can be deduced from the graph?

- There are two divisions.
- There is one DNA duplication.
- DNA content is halved at the end of the whole process.

- (1) and (2) only
- (1) and (3) only
- (2) and (3) only
- (1), (2) and (3)

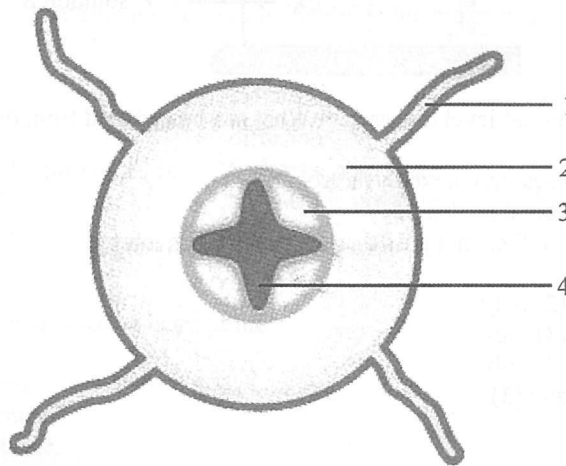
19. Which of the following stages best represent(s) interphase?

- A. P only
- B. Q only
- C. P and Q only
- D. P, Q and R only

20. Sister chromatids separate from one another during the transition from

- A. P to Q.
- B. Q to R.
- C. R to S.
- D. S to T.

Directions: Questions 21 to 23 refer to the diagram below, which shows the cross section of a young dicotyledonous root:



21. Which of the following parts provides rigidity to support the plant?

- A. 1
- B. 2
- C. 3
- D. 4

22. Which of the following parts is responsible for the transport of food?

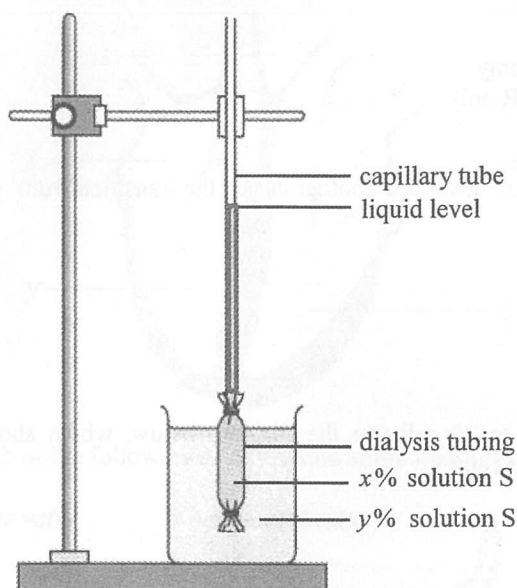
- A. 1
- B. 2
- C. 3
- D. 4

23. Which of the following parts is most likely to have more mitochondria?

- A. 1
- B. 2
- C. 3
- D. 4

Directions:

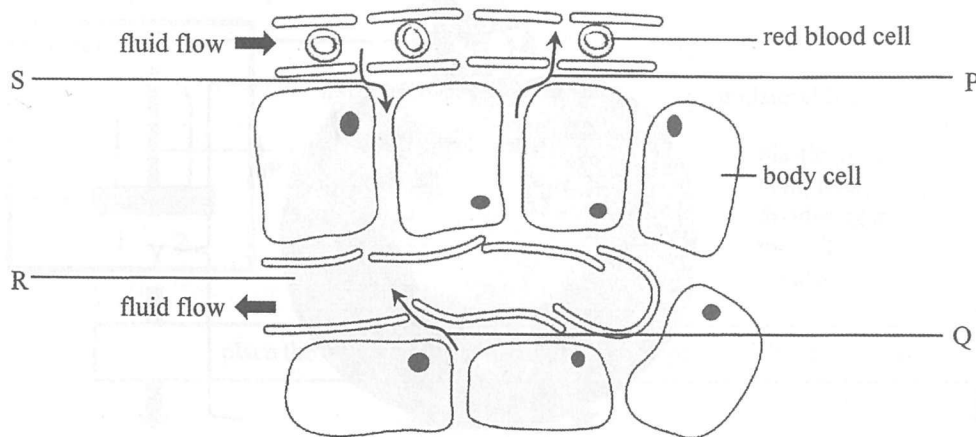
Questions 24 and 25 refer to the diagram below, which shows an experimental set-up for studying osmosis:



24. After three hours, the liquid level has risen. What can be deduced from this result?
- (1) Solution S diffuses into the dialysis tubing.
 - (2) Value of x is larger than that of y .
 - (3) Solute of solution S cannot pass through the dialysis tubing.
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)
25. Which of the following modifications will result in a faster rise in the liquid level?
- A. use a smaller volume of $x\%$ solution S
B. use a larger volume of $y\%$ solution S
C. use a shorter capillary tube
D. use a longer piece of dialysis tubing
26. Which of the following processes in aerobic respiration release NADH?
- (1) glycolysis
 - (2) Krebs cycle
 - (3) oxidative phosphorylation
- A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)
27. Which of the following statements about enzymes is *incorrect*?
- A. Enzymes are produced by cells.
B. Enzymes are denatured at extreme temperatures.
C. There could be more than one specific enzyme to catalyse the same reaction.
D. When an enzyme encounters the same substrates, it always produces the same products.

Directions:

Questions 28 and 29 refer to the diagram below, which shows the cells and the associated vessels in a tissue:



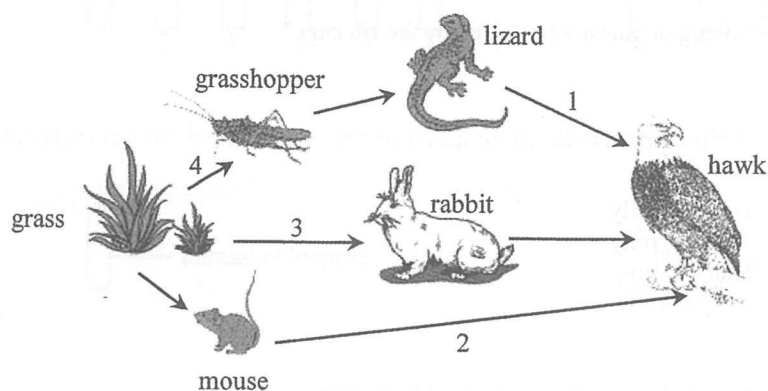
28. Which of the following combinations correctly identifies the major causes of fluid flow represented by P, Q and S?

	<i>P</i>	<i>Q</i>	<i>S</i>
A.	osmosis	diffusion	hydrostatic pressure
B.	hydrostatic pressure	hydrostatic pressure	diffusion
C.	osmosis	hydrostatic pressure	hydrostatic pressure
D.	diffusion	diffusion	osmosis

29. Fluid in R eventually returns to the heart at the

- A. left atrium.
- B. right atrium.
- C. left ventricle.
- D. right ventricle.

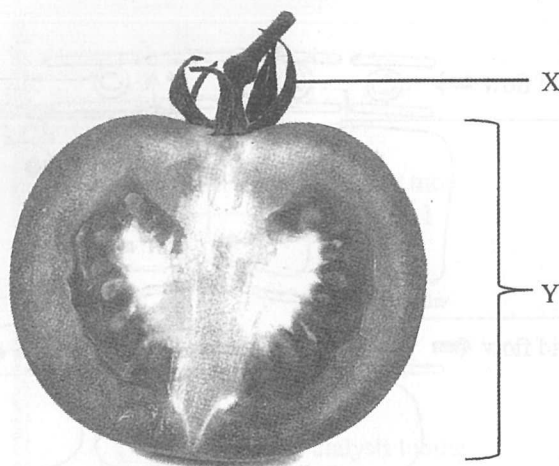
30. The following is a food web found in grassland:



If the energy input for each food chain is the same, the transfer of energy between two trophic levels is the smallest in

- A. 1.
- B. 2.
- C. 3.
- D. 4.

Directions: Questions 31 and 32 refer to the diagram below, which shows the section of a fruit:



31. Which of the following combinations correctly shows the floral parts that develop into structures X and Y?

	X	Y
A.	carpel	petal
B.	carpel	ovary
C.	sepal	petal
D.	sepal	ovary

32. Structure Y serves as

- (1) food store for seed germination.
- (2) attraction to animals for seed dispersal.
- (3) cushion to protect the seed during falling.

- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

33. Which of the following organs are protected by the rib cage?

- (1) lung
- (2) liver
- (3) heart

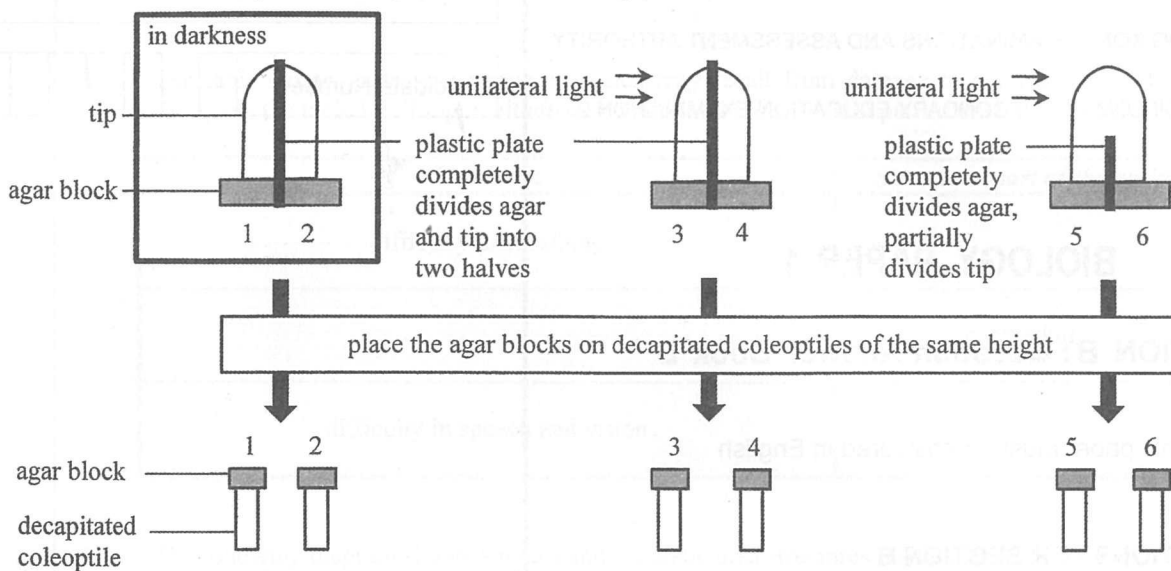
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

34. Symbiotic bacteria found in the root nodules of bean plants are able to convert

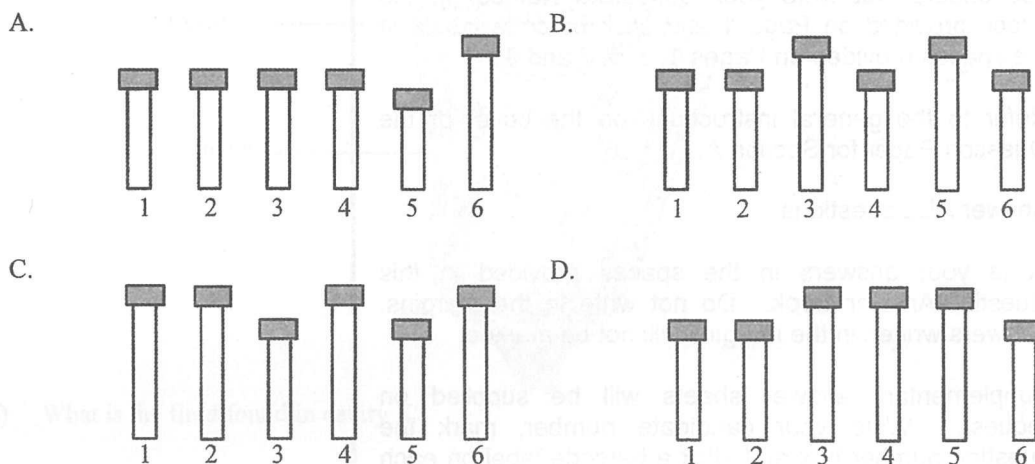
- A. nitrite ions to nitrate ions.
- B. nitrogen gas to nitrate ions.
- C. organic nitrogenous compounds to nitrate ions.
- D. nitrogen gas to ammonium ions.

Directions:

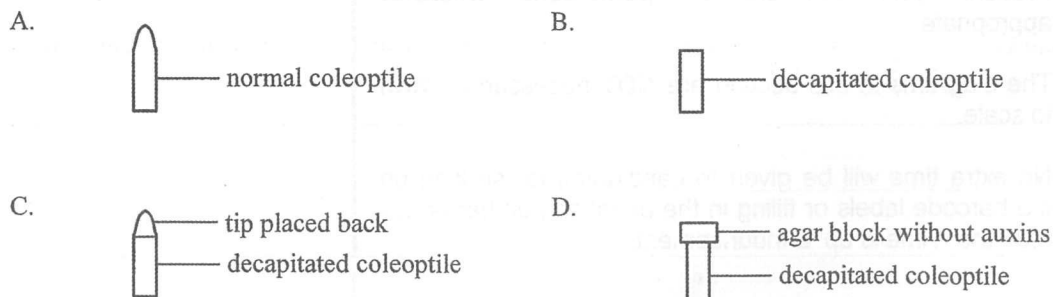
Questions 35 and 36 refer to the diagram below, which shows an experiment that collects auxins from the tip of coleoptiles under different conditions. After that, the agar blocks are placed on decapitated coleoptiles of the same height:



35. Which of the following correctly shows the growth response of the decapitated coleoptiles?



36. Which of the following can be used as a control set-up for the above experiment?



END OF SECTION A

Go on to Question-Answer Book B for questions on Section B